

Data Mining

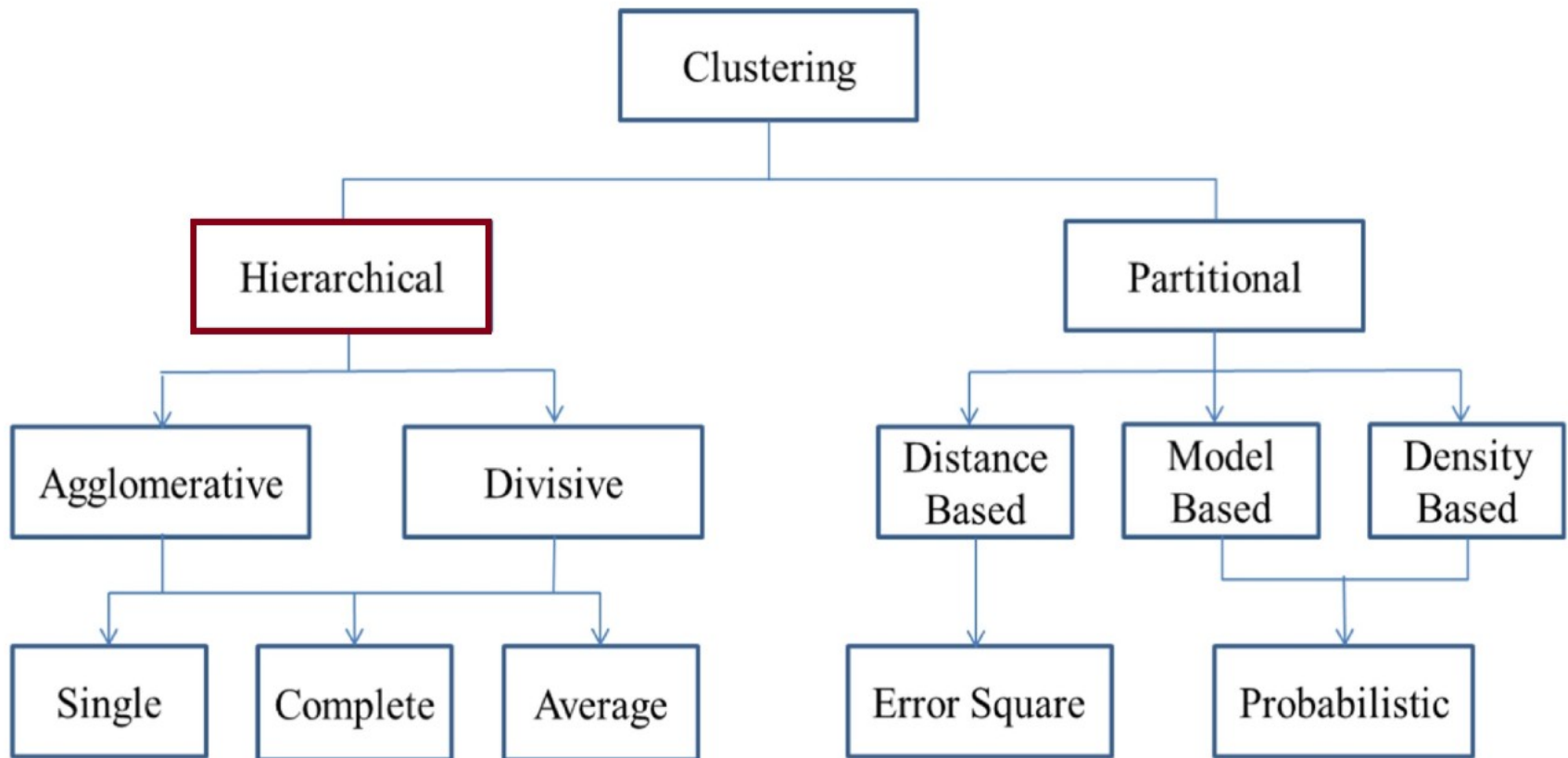
Clustering: Part 2

Mohammed Brahim & Sami Belkacem

Outline

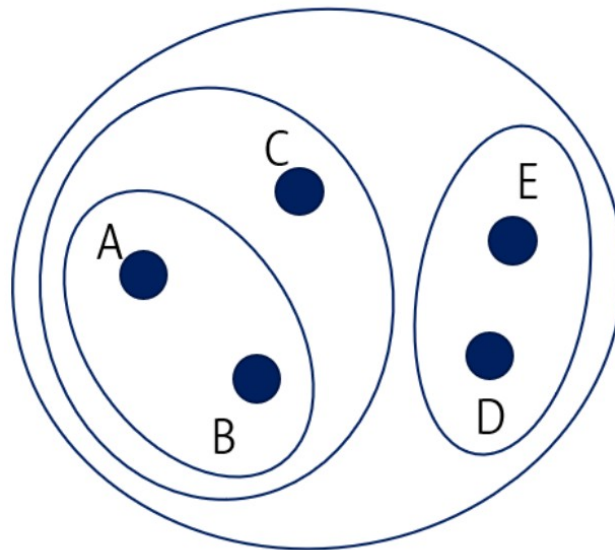
- ☐ Overview of Clustering
- ☐ Major Clustering Approaches
 - ☐ K-means Clustering
 - ☐ Hierarchical Clustering
 - ☐ DBSCAN Clustering
- ☐ Cluster Evaluation

Major Clustering Approaches



Hierarchical Clustering

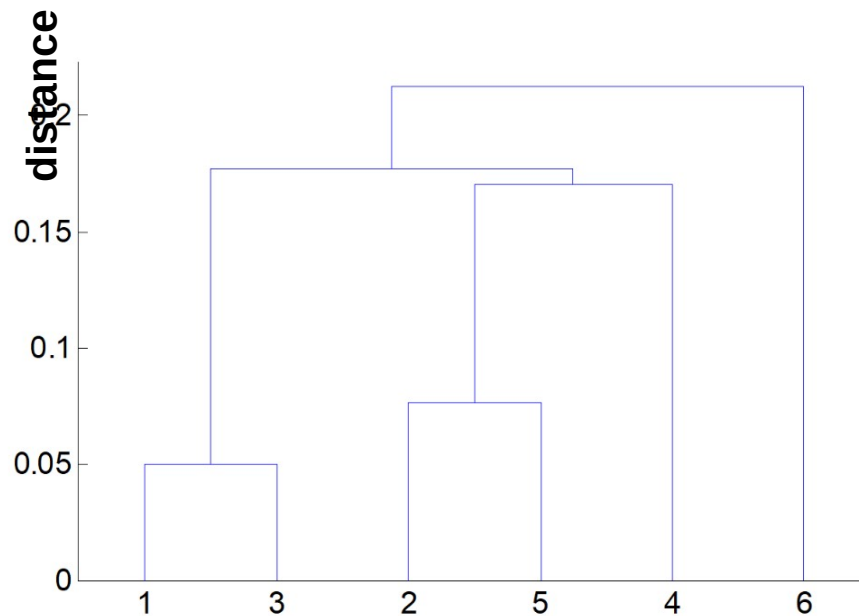
- Hierarchical Clustering produce a set of nested-clusters.
- It does not have to assume any particular number of clusters.
- It may correspond to meaningful taxonomies (e.g., biological taxonomy, animal kingdom, phylogeny reconstruction, ...).



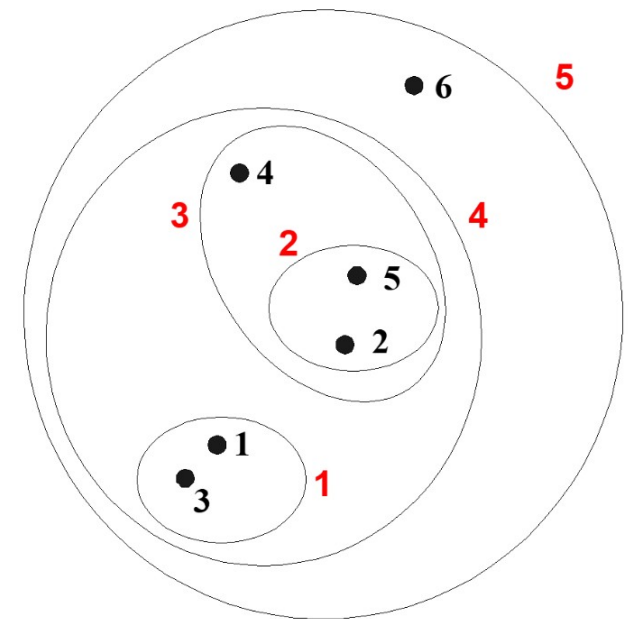
**Nested clusters
of 5 data points**

Hierarchical Clustering

- The set of nested clusters can be organized as a hierarchical tree.
- The hierarchical tree of clusters is called a dendrogram, which records the sequences of merges or splits
- Different clustering of the data can be obtained by cutting the dendrogram at the desired level, then each connected component forms a cluster



Dendrogram



5 nested clusters of 6 data points

Types of Hierarchical Clustering

■ Agglomerative:

- Start with the points as individual clusters
- At each step, merge the **closest** pair of clusters
- Until only one cluster (or ***k*** clusters) left
- **Popular algorithm:** AGNES (Agglomerative Nesting)

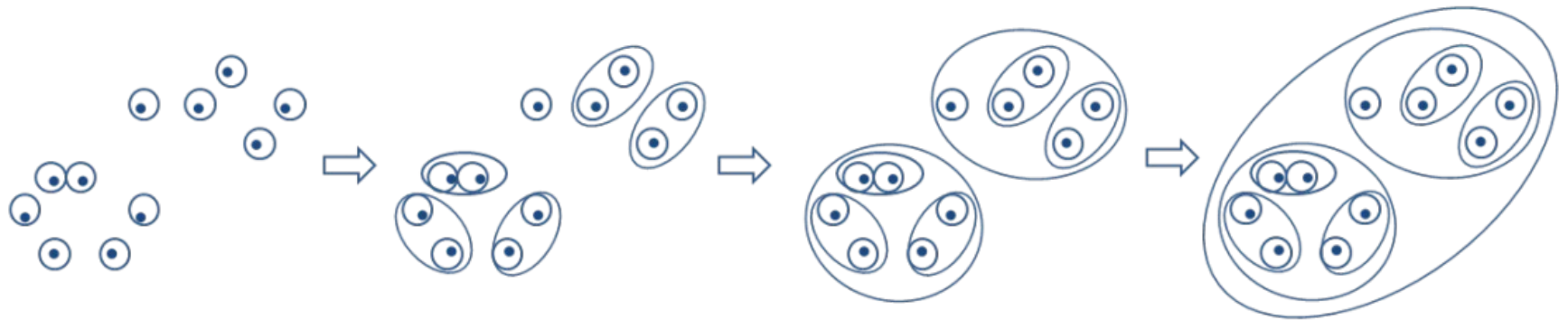
■ Divisive:

- Start with one, all-inclusive cluster
- At each step, split the **least cohesive** clusters
- Until each cluster contains one point (or there are ***k*** clusters)
- **Popular algorithm:** DIANA (Divisive Analysis)

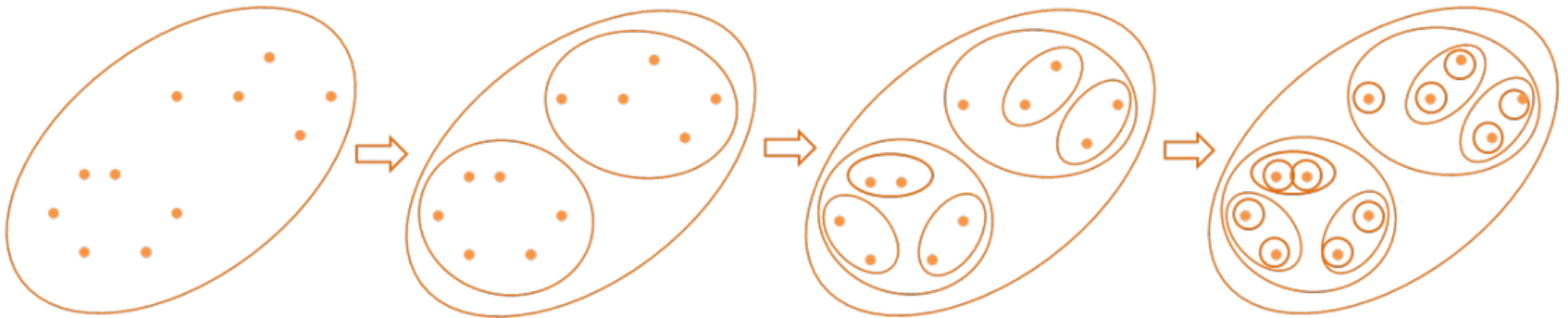
- Hierarchical algorithms use a proximity matrix (similarity or distance)
 - Merge or split one cluster at a time

Agglomerative vs Divisive

Agglomerative Hierarchical Clustering



Divisive Hierarchical Clustering

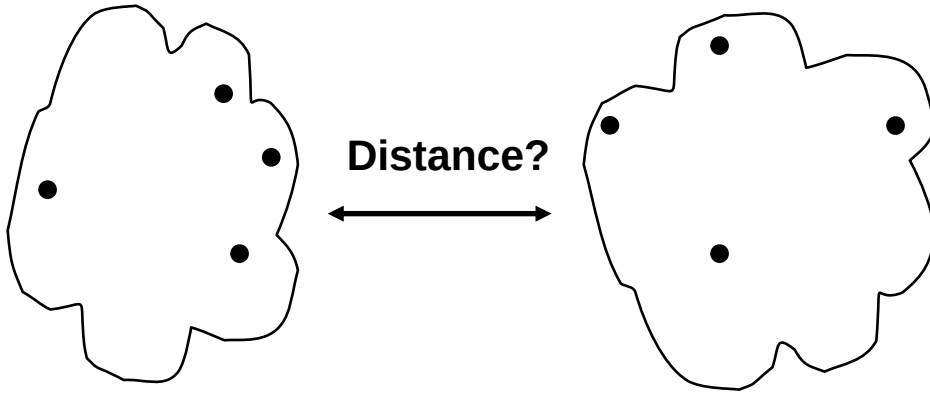


Agglomerative Clustering Algorithm

- **Key Idea:** Successively merge the closest clusters
- **Basic algorithm:**
 1. Compute the proximity matrix for all data points
 2. Let each data point be a cluster
 3. **Repeat**
 4. **Merge the two closest clusters**
 5. Update the proximity matrix
 6. **Until** only a single cluster remains (or k clusters left)
- Key operation is the computation of the proximity of two clusters:
 - Different approaches to defining the **distance between clusters** distinguish the different algorithms (Min, Max, etc.)

How to measure the distance
between two clusters?

How to Define Inter-Cluster Distance

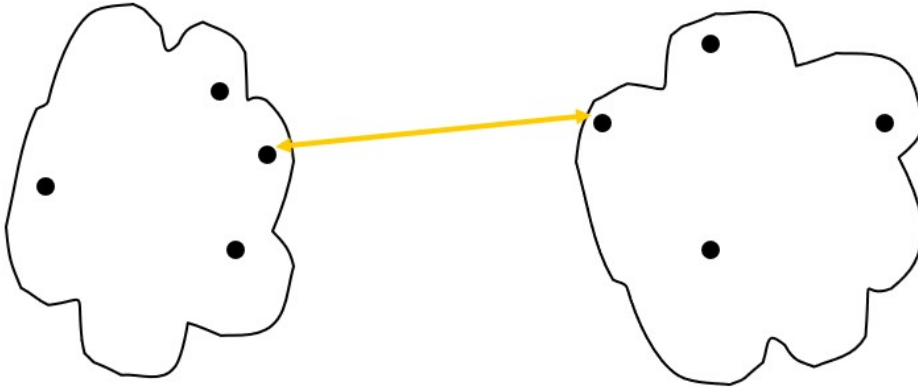


1. MIN
2. MAX
3. Group Average
4. Distance Between Centroids

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

How to Define Inter-Cluster Distance



1. MIN

2. MAX

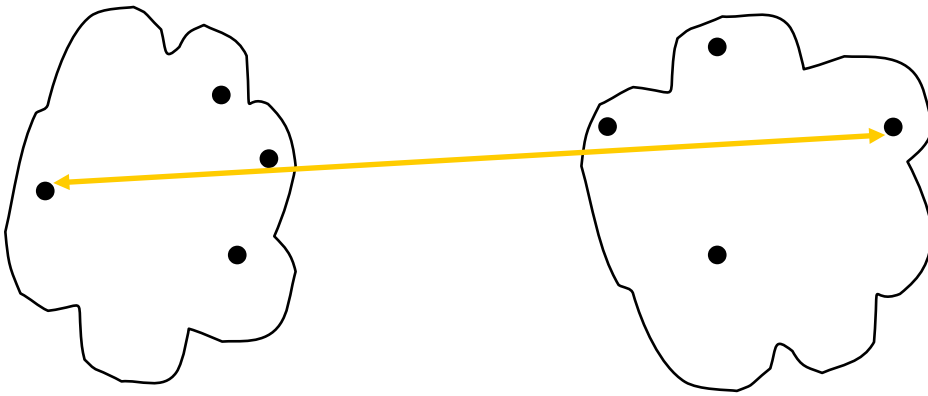
3. Group Average

4. Distance Between Centroids

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

How to Define Inter-Cluster Distance



1. MIN

2. MAX

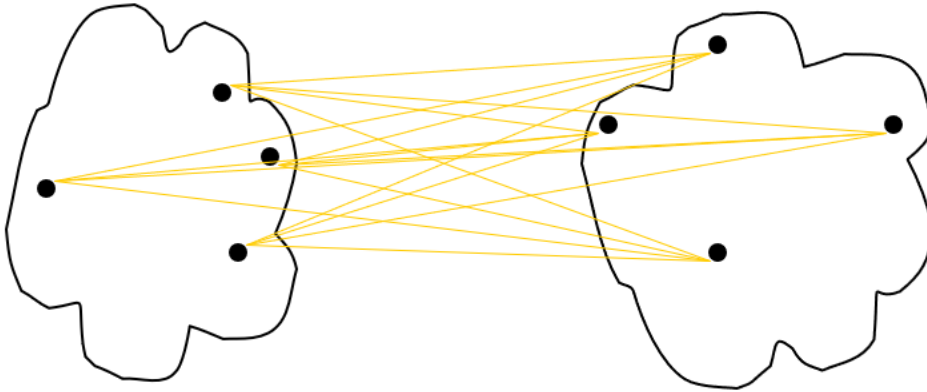
3. Group Average

4. Distance Between Centroids

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

How to Define Inter-Cluster Distance



1. MIN

2. MAX

3. Group Average

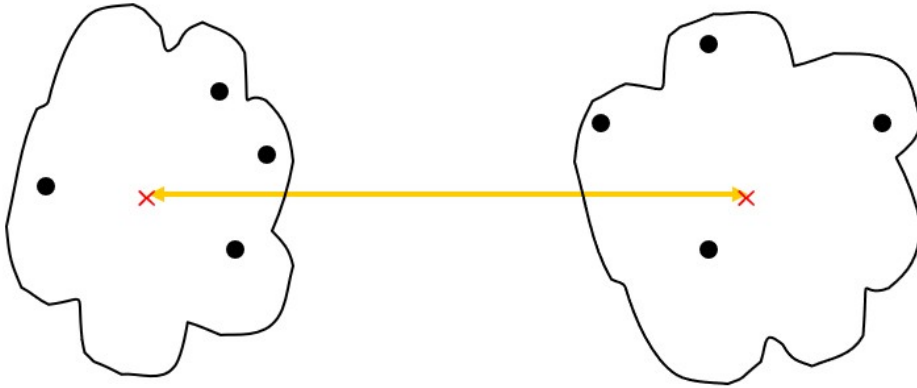
4. Distance Between Centroids

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

$$\text{proximity}(\text{Cluster}_i, \text{Cluster}_j) = \frac{\sum_{\substack{p_i \in \text{Cluster}_i \\ p_j \in \text{Cluster}_j}} \text{proximity}(p_i, p_j)}{|\text{Cluster}_i| \times |\text{Cluster}_j|}$$

How to Define Inter-Cluster Distance



1. MIN

2. MAX

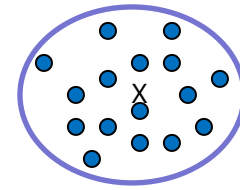
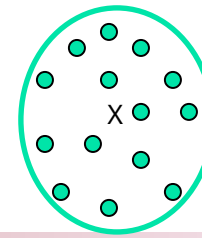
3. Group Average

4. Distance Between Centroids

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

Inter-Cluster Distance

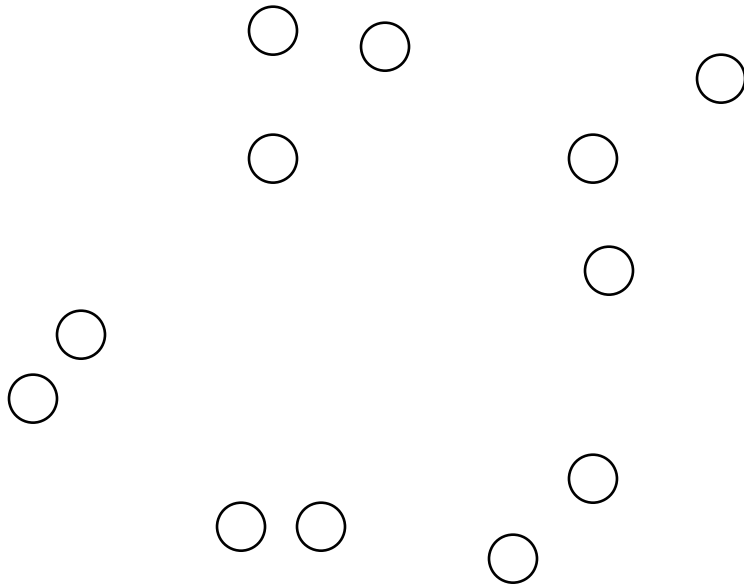


- 1. Min (Single link):** smallest distance between an element in one cluster and an element in the other, $\text{dist}(K_i, K_j) = \min(t_{ip}, t_{jq})$
- 2. Max (Complete link):** largest distance between an element in one cluster and an element in the other, $\text{dist}(K_i, K_j) = \max(t_{ip}, t_{jq})$
- 3. Group Average:** avg distance between an element in one cluster and an element in the other, $\text{dist}(K_i, K_j) = \text{avg}(t_{ip}, t_{jq})$
- 4. Centroid:** distance between the centroids of two clusters $\text{dist}(K_i, K_j) = \text{dist}(C_i, C_j)$

Now that we've understood how to measure the distance between two clusters, let's go back to the steps of the Agglomerative Clustering algorithm.

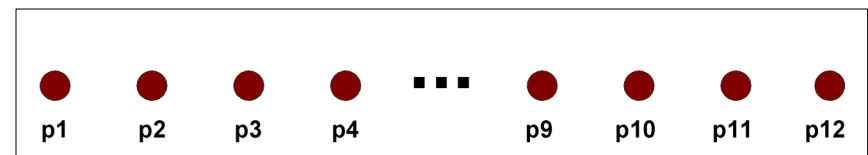
Agglomerative clustering: Steps 1 and 2

- Start with clusters of individual points and a proximity matrix



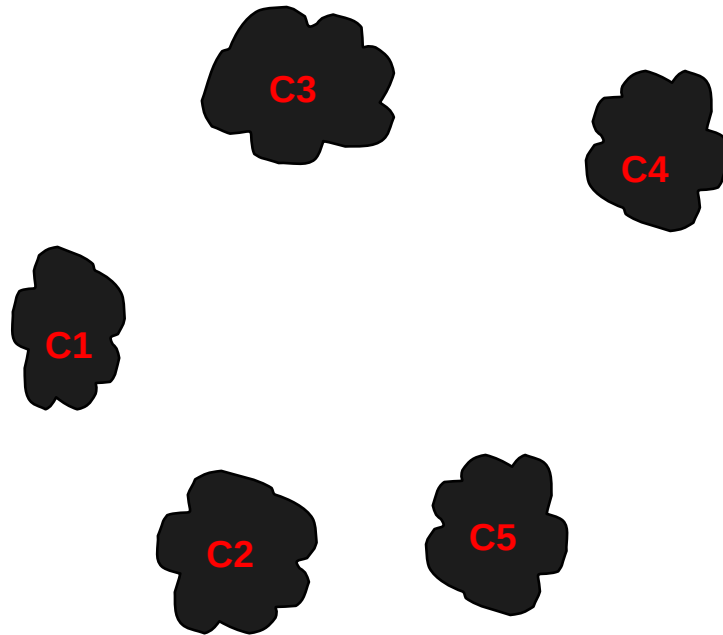
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix



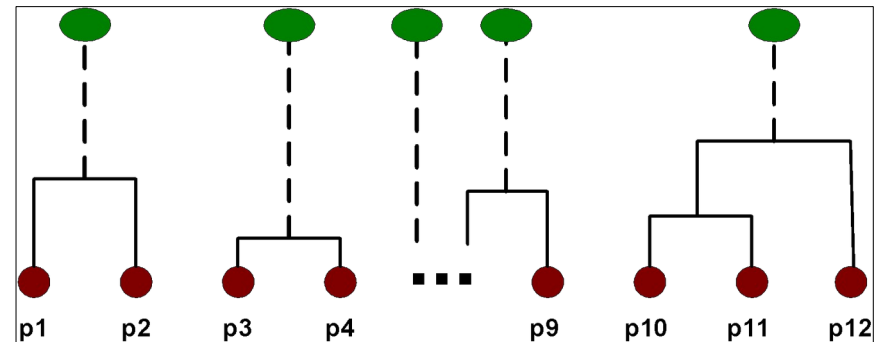
Intermediate Situation

- After some merging steps, we have some clusters



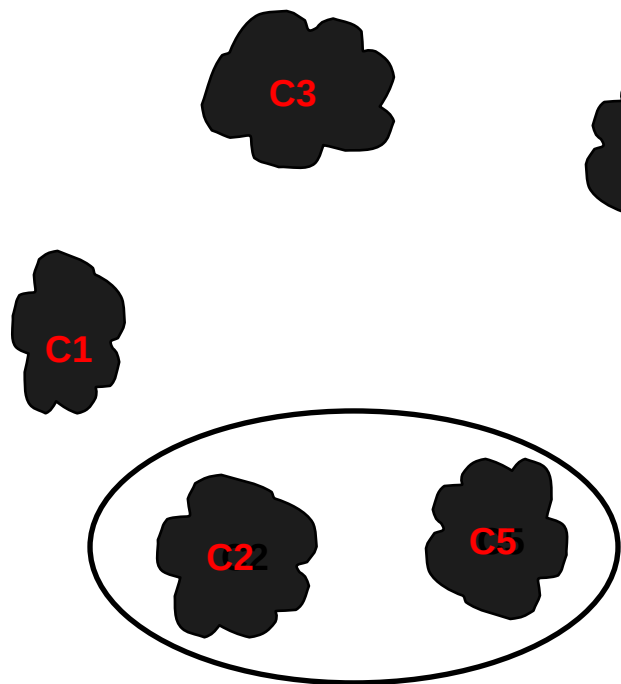
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



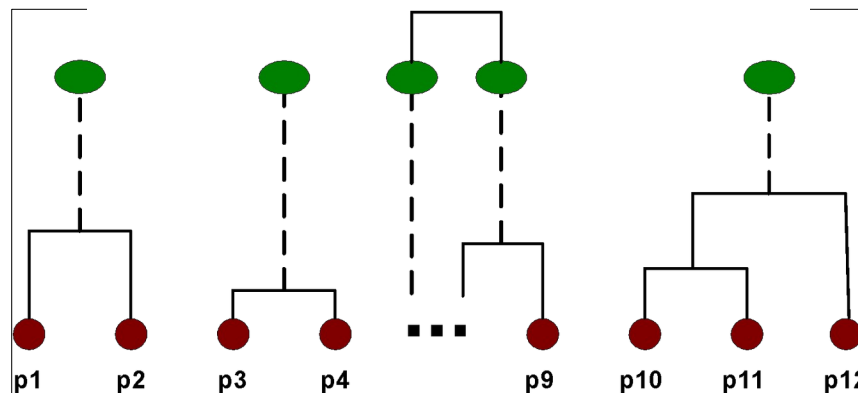
Step 4

- **Merge** the two closest clusters (C2 and C5) and **update** the matrix



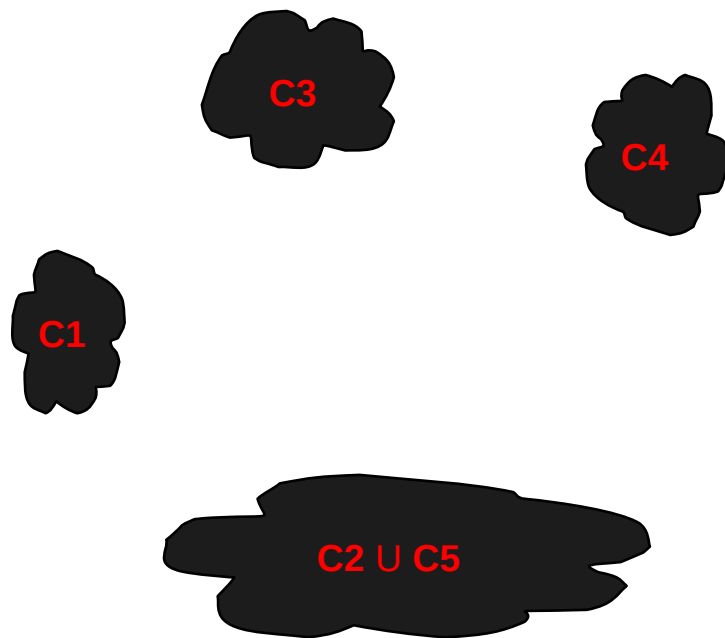
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



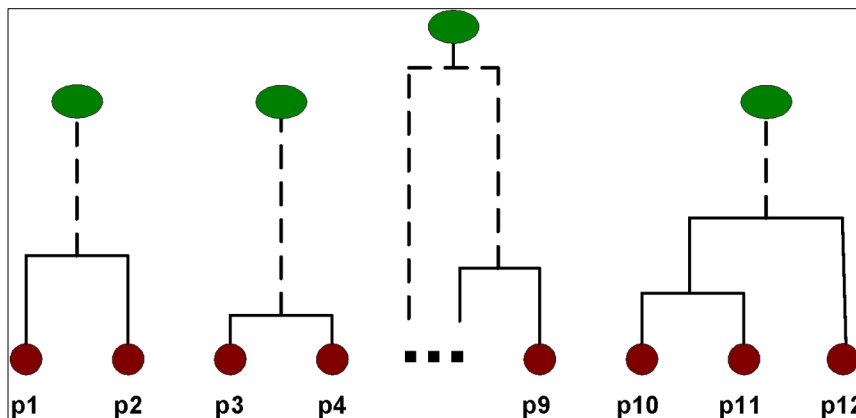
Step 5

■ Now, the question is “**how do we update the proximity matrix?**”



		C2 U			
		C1	C5	C3	C4
C2 U C5	C1		?		
		?	?	?	?
	C3		?		
	C4		?		

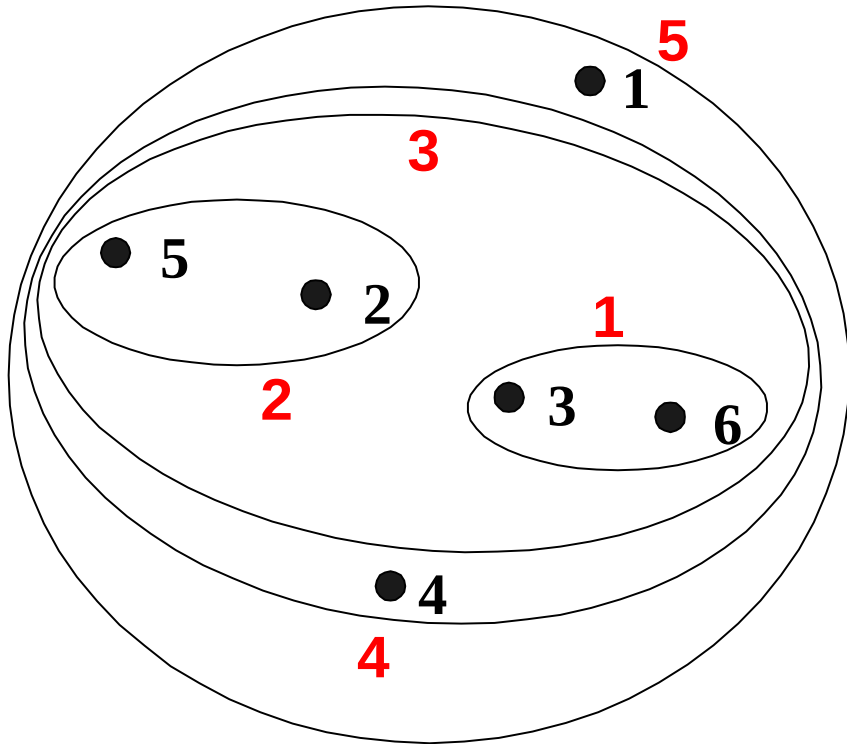
Proximity Matrix



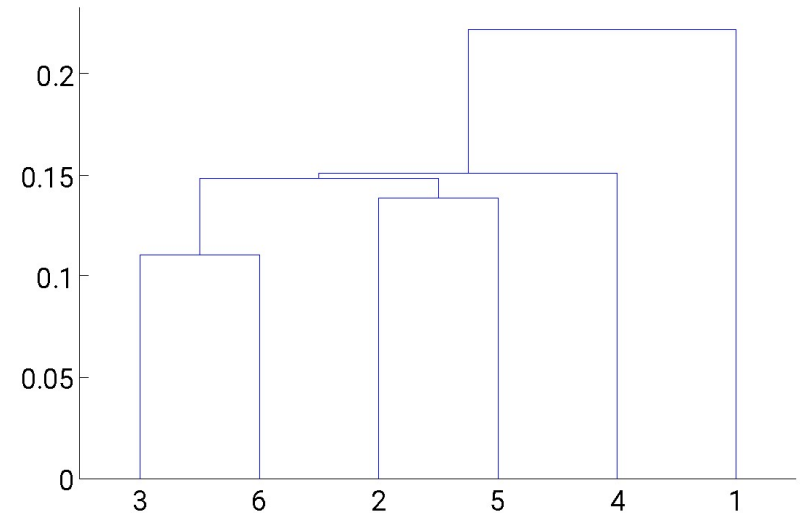
Answer: we update the proximity matrix using the different approaches to defining the distance between clusters (Min, Max, etc.)

Note: to compute the distance between an individual data point and a cluster, we consider that data point itself as a cluster

Hierarchical Clustering: MIN

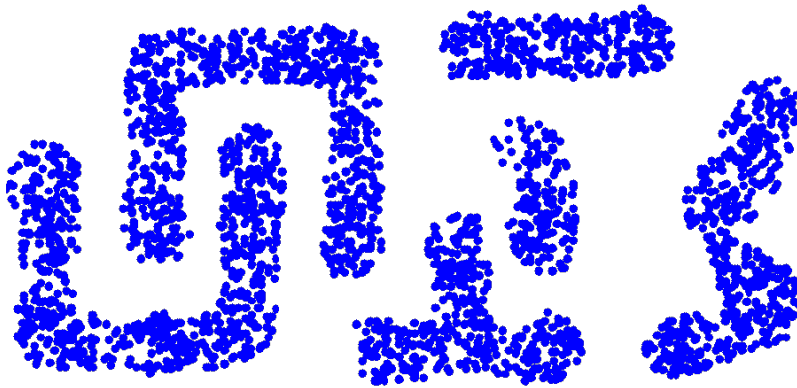


Nested Clusters

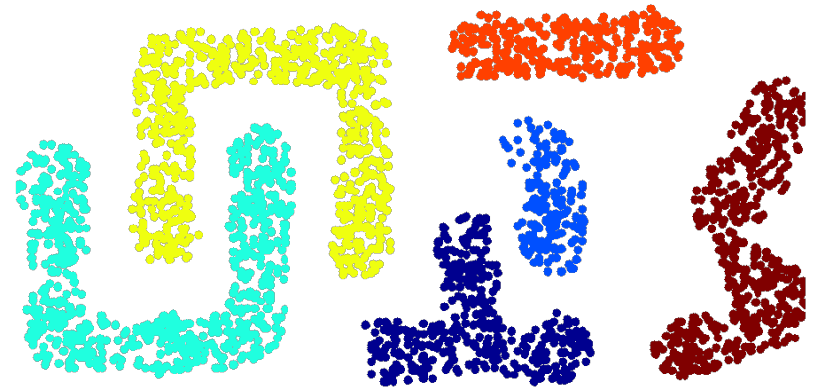


Dendrogram

Strength of MIN



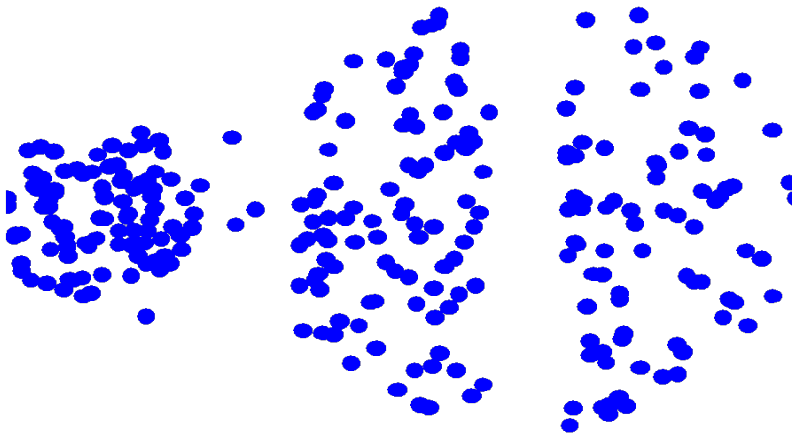
Original Points



Six Clusters

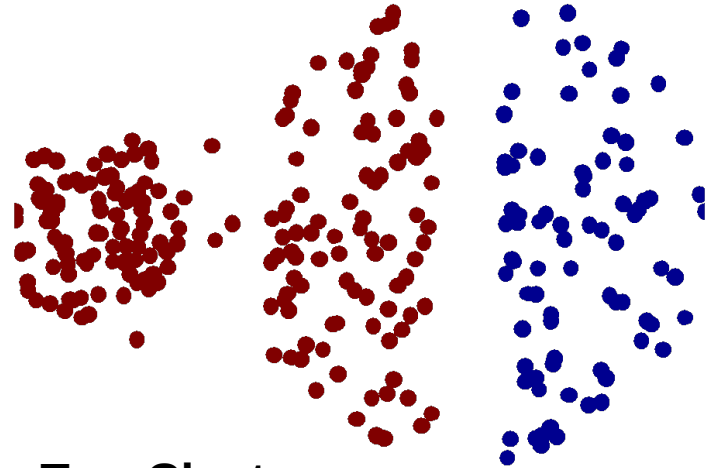
- o It detects clusters of any shape by focusing only on the nearest points between clusters, **ignoring overall shape**.
- o Captures irregularly shaped clusters effectively without assuming specific geometrical forms like **elliptical shapes**.

Limitations of MIN

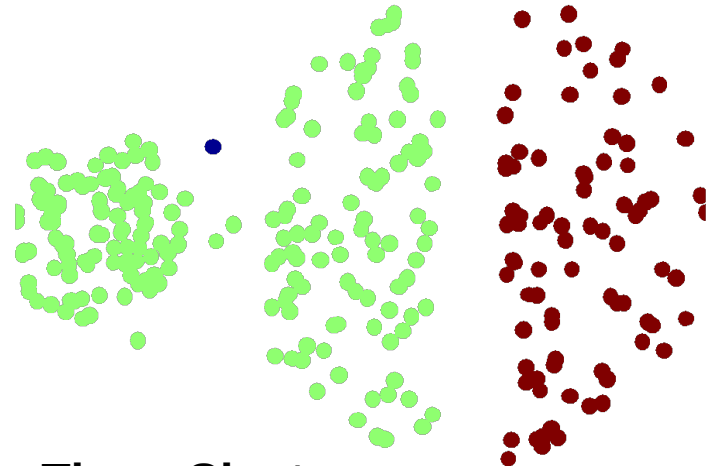


Original Points

- **Chaining effect:** Merges two clusters due to closely paired points, leading to a chain of combined clusters.
- **Noise sensitivity:** A single point can alter the cluster's shape.

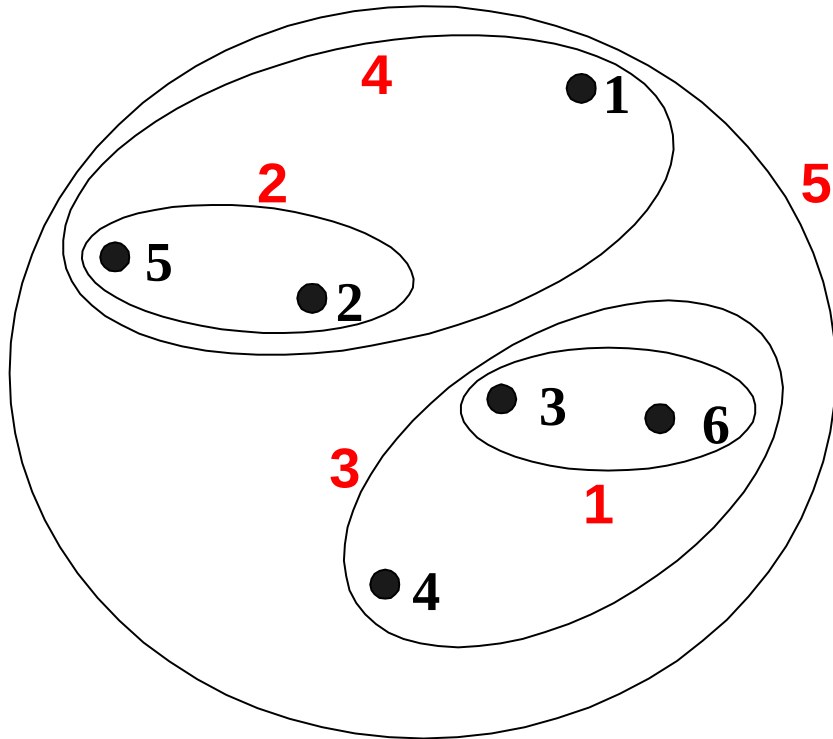


Two Clusters

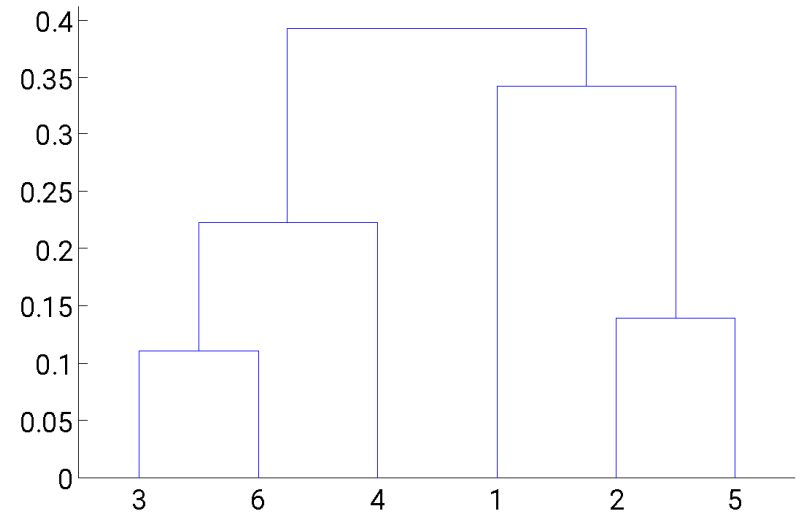


Three Clusters

Hierarchical Clustering: MAX

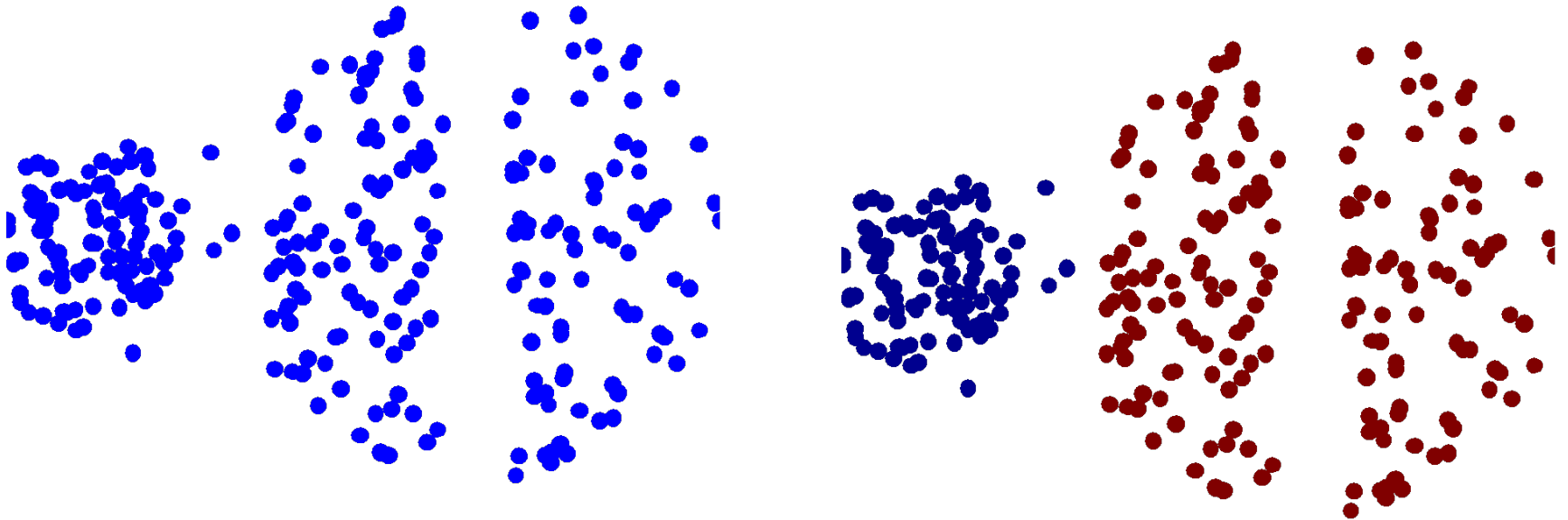


Nested Clusters



Dendrogram

Strength of MAX

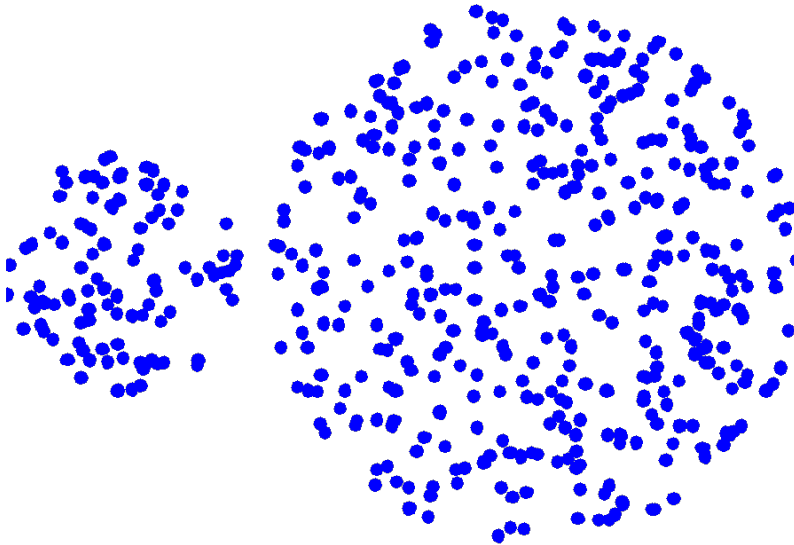


Original Points

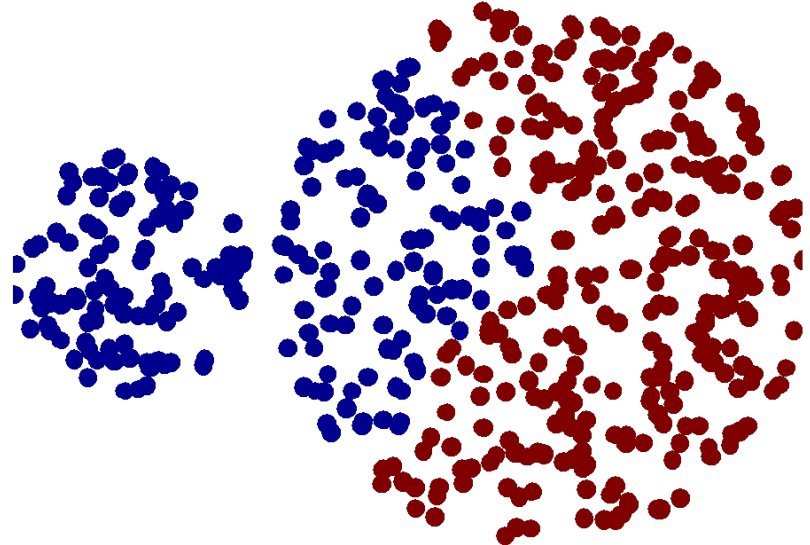
Two Clusters

- **Robustness to Noise:** Less affected by noise because it looks at the farthest points between clusters, forming compact groups less likely to be influenced by outliers.

Limitations of MAX



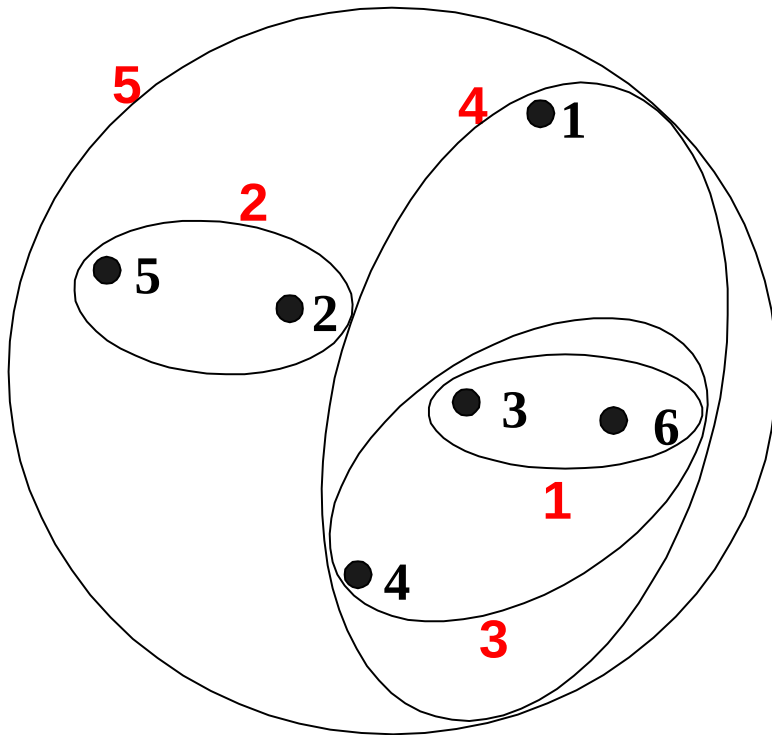
Original Points



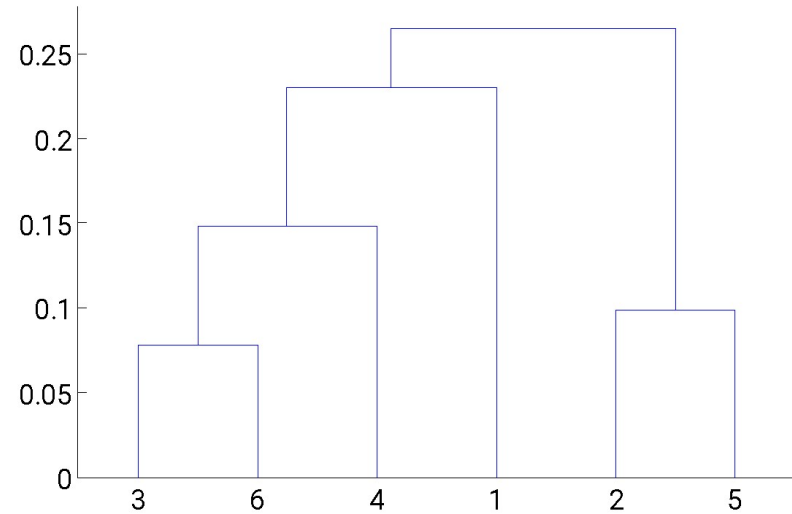
Two Clusters

- o Tends to break large clusters into smaller, more distinct ones.
- o Biased towards globular clusters

Hierarchical Clustering: Group Average



Nested Clusters



Dendrogram

Hierarchical Clustering: Group Average

- Compromise between Single and Complete Link
- **Strengths**
 - Averaging reduces the influence of noisy data points
- **Limitations**
 - Biased towards globular clusters because the average distance favors clusters with compact, closely located points

Hierarchical Clustering: Space and Time Complexity

- N is the number of data points or objects.
- Space: $O(N^2)$
 - $O(N^2)$ because the proximity matrix has N^2 entries for distances between N points.
- Time: $O(N^3)$
 - Find the min distance of the matrix $O(N^2) * N$ iterations $\Rightarrow O(N^3)$
 - Complexity can be reduced to $O(N^2 \log(N))$
 - Accelerate finding the minimum using a heap

Strength of Hierarchical Clustering

- Do not have to assume any particular number of clusters.
- Any desired number of clusters can be obtained by 'cutting' the dendrogram at the proper level.
- Hierarchical clusters may correspond to meaningful taxonomies:
 - Example in biological sciences (e.g., animal kingdom, phylogeny reconstruction, ...)

Weakness of Hierarchical Clustering

- Once a decision is made to combine two clusters, it cannot be undone
- Do not scale well: time complexity of $O(n^3)$, n is the number of objects
- No global objective function is directly minimized
- Different schemes have problems with one or more of the following:
 - Sensitivity to noise
 - Difficulty handling clusters of different sizes and non-globular shapes
 - Breaking large clusters

Improvements: Integration of hierarchical and distance-based clustering

- Example of Algorithms: BIRCH, CHAMELEON